

Module A: Least-squares view (12.1) + minimum-norm exact solution (12.3)

Big picture All these formulas come from *projection geometry*: least squares projects b onto $\mathcal{R}(A)$ to **find the best solution among all imperfect ones**; minimum-norm solutions pick the solution in $\mathcal{R}(A^\top)$ (orthogonal to $\mathcal{N}(A)$) to **find the simplest solution among the equivalently effective ones**.

12.1 Least squares for $Ax \approx b$.

Given $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$:

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2.$$

Normal equations. Any minimizer satisfies

$$A^\top(Ax - b) = 0 \quad \Leftrightarrow \quad A^\top Ax = A^\top b.$$

Full column rank ($m \geq n$, $\text{rank}(A) = n$). Then $A^\top A \succ 0$ and the minimizer is unique:

$$x_{\text{LS}} = (A^\top A)^{-1} A^\top b.$$

Geometric intuition. The residual $r = b - Ax_{\text{LS}}$ is orthogonal to $\mathcal{R}(A)$, i.e., $r \perp \mathcal{R}(A)$.

12.3 Minimum-norm solution of $Ax = b$ (underdetermined).

Assume $A \in \mathbb{R}^{m \times n}$, $m \leq n$, $\text{rank}(A) = m$ (full row rank) and $b \in \mathbb{R}^m$. There are infinitely many x with $Ax = b$; we choose the closest to the origin:

$$\min_{x \in \mathbb{R}^n} \|x\| \quad \text{s.t.} \quad Ax = b.$$

Key theorem (minimum-norm formula). The unique minimizer is

$$x^* = A^\top (AA^\top)^{-1} b.$$

Intuition. Any solution can be decomposed as $x = x_{\mathcal{R}(A^\top)} + x_{\mathcal{N}(A)}$. The constraint $Ax = b$ fixes the $\mathcal{R}(A^\top)$ component; the minimum norm sets the nullspace component to 0, hence $x^* \in \mathcal{R}(A^\top)$.

! Remarks

- $\mathcal{R}(A^\top)$ could be understood as the **effective domain** of A .
- Two “projections”:** $A(A^\top A)^{-1} A^\top$ is the projector onto $\mathcal{R}(A)$ (when $\text{rank}(A) = n$); $A^\top (AA^\top)^{-1} A$ is the projector onto $\mathcal{R}(A^\top)$ (when $\text{rank}(A) = m$).

📖 Practice: Textbook 12.18 (projection onto $\mathcal{R}(A)$)

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $m \geq n$, and $\text{rank}(A) = n$. Consider

$$\frac{1}{2} x^\top x - x^\top b \quad \text{subject to} \quad x \in \mathcal{R}(A),$$

where $\mathcal{R}(A)$ denotes the range of A . Derive an expression for the global minimizer in terms of A and b .

Answer: Parameterize the constraint by $x = Az$ with $z \in \mathbb{R}^n$:

$$\phi(z) = \frac{1}{2} \|Az\|^2 - (Az)^\top b = \frac{1}{2} z^\top (A^\top A) z - z^\top A^\top b.$$

Since $\text{rank}(A) = n$, $A^\top A \succ 0$, so the unique minimizer satisfies

$$\nabla \phi(z) = A^\top Az - A^\top b = 0 \Rightarrow z^* = (A^\top A)^{-1} A^\top b.$$

Thus

$$x^* = Az^* = A(A^\top A)^{-1} A^\top b.$$

Equivalently, $x^* = P_{\mathcal{R}(A)} b$ is the orthogonal projection of b onto $\mathcal{R}(A)$.



Module B: Kaczmarz's algorithm = cyclic row projections for Min Norm Solution (12.4)

We want to solve a consistent system $Ax = b$ with $A \in \mathbb{R}^{m \times n}$, $m \leq n$, $\text{rank}(A) = m$. Write the rows of A as $a_1^\top, \dots, a_m^\top$ and $b = (b_1, \dots, b_m)^\top$.

Algorithm (cyclic row updates). Choose $0 < \mu < 2$ and an initial point $x^{(0)}$. For $k = 0, 1, 2, \dots$, pick the row index

$$j = R(k) + 1 \quad (\text{i.e., } j = (k \bmod m) + 1),$$

and update

$$x^{(k+1)} = x^{(k)} + \underbrace{\frac{b_j - a_j^\top x^{(k)}}{a_j^\top a_j} a_j}_{\text{project direction orthogonally towards } b_j = a_j^\top x^{(k)}}.$$

Geometric intuition. The equation $a_j^\top x = b_j$ defines a hyperplane where this entry of data constraint is satisfied / loss is 0. When $\mu = 1$, the update is the *orthogonal projection* of $x^{(k)}$ onto that hyperplane. Cyclic Kaczmarz repeatedly projects onto each row-hyperplane.

Convergence to Minimum Norm Solution. If $x^{(0)} = 0$, then $x^{(k)}$ converges to the *minimum-norm solution*

$$x^\star = A^\top (AA^\top)^{-1} b.$$

More generally, if $x^{(0)} \neq 0$, it converges to the point in $\{x : Ax = b\}$ closest to $x^{(0)}$.



Remarks

- μ is a **relaxation** parameter: $\mu = 1$ is pure projection; $0 < \mu < 2$ is required for the standard convergence proof.
- The method is **row-scaled**: the factor $1/(a_j^\top a_j)$ normalizes the step by the row norm.



Practice: Textbook 12.28 (projection property when $\mu = 1$)

Show that in Kaczmarz's algorithm, if $\mu = 1$, then

$$b_{R(k)+1} - a_{R(k)+1}^\top x^{(k+1)} = 0 \quad \text{for each } k.$$

Answer: Let $j = R(k) + 1$ and set $\mu = 1$. The update is

$$x^{(k+1)} = x^{(k)} + \frac{b_j - a_j^\top x^{(k)}}{a_j^\top a_j} a_j.$$

Multiply by $a_j^\top$:

$$a_j^\top x^{(k+1)} = a_j^\top x^{(k)} + \frac{b_j - a_j^\top x^{(k)}}{a_j^\top a_j} a_j^\top a_j = a_j^\top x^{(k)} + (b_j - a_j^\top x^{(k)}) = b_j.$$

Hence $b_j - a_j^\top x^{(k+1)} = 0$, i.e. $b_{R(k)+1} - a_{R(k)+1}^\top x^{(k+1)} = 0$.